

Paired Certification of Simultaneous-Inference Gains under Scarce Reference Verification

Core-method technical note for the JRSSB revision

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Abstract

This note develops the first technical component of the proposed revision of *Surrogate-Assisted Local Conditional Distribution Inference with Scarce Reference Outcomes*. An explicit CDF example separates integrated-variance minimization from simultaneous-band minimization. We construct a borrowing rule by lower-bounding a paired Gaussian critical-value gain over a common joint-covariance confidence set. The resulting certificate controls the probability of selecting a harmful borrowing weight and satisfies an oracle inequality without requiring a unique minimizer. On a fixed nondegenerate threshold grid, the error in the paired gain is proportional to the borrowing weight times the covariance-estimation error. An explicit observable covariance confidence ball is provided. A binary local submodel yields a matching signal-detection obstruction and certificate: the signal scale is the inverse square root of the local verified count, whereas the optimal critical-value gain has its square as the relevant scale. The attached code reproduces the separation example, checks the paired cancellation, and verifies the submodel certificate.

1 Research target and relationship to the submitted manuscript

The supplied 36-page manuscript defines the local target and AIPW signal in Section 2.1, chooses a common borrowing weight by integrated augmentation variance in Section 2.2, and establishes raw-process simultaneous bands in Section 3.3. Its Section 3.4 gives the covariance cost of augmentation error. These components are retained. The added object is the *gain in a specified simultaneous-inference criterion*, together with the information required to certify that gain.

Several nearby contributions already cover components of this proposal. PPI++ develops efficient adaptive borrowing for finite-dimensional parameters (Angelopoulos et al., 2024). Conditional PPI develops local conditional inference and nonasymptotic and minimax results (Sui et al., 2026); its revised version dated 29 August 2026 was checked. Generation-powered inference develops an optimal functional borrowing operator and simultaneous bands (Zhang and Li, 2026). Multi-source PPI directly optimizes asymptotic confidence-region volume (Li et al., 2026). Finite-sample costs of estimating PPI++ weights are studied by Mani et al. (2025). Learning a candidate and then validating a risk claim is also an established idea (Angelopoulos et al., 2021).

Accordingly, neither AIPW augmentation, optimization of a confidence-region size, nor a generic lower-confidence-bound decision is treated here as a new principle. The technical object pursued here is a *paired, local, distribution-level gain certificate* whose uncertainty vanishes at the anchor and whose weak-signal behavior can be related to local verified information. An operator that dominates all competing covariance operators already improves probabilities of centered convex Gaussian sets. The separation example below concerns a fitted scalar path which need not admit such a dominating element; it does not contradict operator-efficiency results.

2 Local scores and a common covariance representation

Condition on an independent training sample. The fitted candidates $q_y(X)$ and $m_y(W)$ are now fixed, bounded in $[0, 1]$, and need not be correctly specified. Write

$$D_y = m_y - q_y, \quad a_{\lambda,y} = q_y + \lambda D_y, \quad 0 \leq \lambda \leq 1.$$

Let $O = (W, R, RY)$, $R \perp Y \mid W$, and let the known verification probability satisfy $\pi(W) = pe(W)$, with $p > 0$ and $e(W) \geq c_e > 0$. For $Z_y = \mathbf{1}\{Y \leq y\}$,

$$U_{\lambda,y} = q_y + \frac{R}{\pi}(Z_y - q_y) + \lambda \left(1 - \frac{R}{\pi}\right) D_y = U_{0,y} + \lambda B_y, \quad (1)$$

$$B_y = \left(1 - \frac{R}{\pi}\right) D_y. \quad (2)$$

Both $\mathbb{E}(B_y \mid W) = 0$ and $\mathbb{E}(U_{\lambda,y} \mid W) = \mathbb{E}(Z_y \mid W)$ hold exactly.

Fix x_0 , a nonnegative bounded kernel $K_h(X - x_0)$, and $v_h = h^d$. Put

$$\mu_h = \mathbb{E}K_h > 0, \quad f_h = \mu_h/v_h, \quad F_h(y) = \mathbb{E}(K_h Z_y)/\mu_h.$$

Use a fixed threshold grid $y_1 < \dots < y_J$. Define the $2J$ -vector

$$\Psi(O) = \sqrt{p/v_h} K_h \begin{pmatrix} U_{0,y_1} - F_h(y_1) \\ \vdots \\ U_{0,y_J} - F_h(y_J) \\ B_{y_1} \\ \vdots \\ B_{y_J} \end{pmatrix}, \quad \Omega = \mathbb{E}(\Psi\Psi^\top). \quad (3)$$

It has mean zero. With $H_\lambda = (I_J, \lambda I_J)$,

$$\Sigma_\lambda = H_\lambda \Omega H_\lambda^\top = \Omega_{00} + \lambda(\Omega_{01} + \Omega_{10}) + \lambda^2 \Omega_{11}. \quad (4)$$

The usual covariance for the $\sqrt{np h^d}$ -scaled kernel-ratio estimator is Σ_λ/f_h^2 . The positive scale f_h^{-1} is common to all weights and therefore cancels in weight selection and the sign of a gain.

For an independent evaluation sample of size n_E ,

$$\hat{F}_{E,\lambda}(y) = \frac{\mathbb{P}_E K_h U_{\lambda,y}}{\mathbb{P}_E K_h}, \quad \sqrt{n_E p v_h}(\hat{F}_{E,\lambda} - F_h) = \frac{n_E^{-1/2} \sum_{i \in E} H_\lambda \Psi(O_i)}{\hat{f}_{h,E}}, \quad (5)$$

where $\hat{f}_{h,E} = \mathbb{P}_E K_h/v_h$. This identity is exact whenever the denominator is positive.

Let $G_\lambda \sim N(0, \Sigma_\lambda)$ and define

$$c_\Omega(\lambda) = \inf\{t : \mathbb{P}(\|G_\lambda\|_\infty \leq t) \geq 1 - \alpha\}, \quad g_\Omega(\lambda) = c_\Omega(0) - c_\Omega(\lambda). \quad (6)$$

The raw band has common half-width $c_\Omega(\lambda)/(f_h \sqrt{n_E p v_h})$ at population calibration. A fixed positive shape w_j can be accommodated by dividing the j th coordinate of both score blocks by w_j before defining Ω . Shapes depending on λ are excluded from this comparison.

3 A verifiable separation between two efficiency criteria

Let $S \sim \text{Bernoulli}(0.2)$ and

$Y \mid S = 0 \in \{1, 2\}$ with equal probabilities, $Y \mid S = 1 \in \{0, 2\}$ with equal probabilities.

Take thresholds $(0, 1)$, $q = (0.1, 0.5)$, $p = 0.1$, independent verification, and the legitimate candidate CDF values

$$m(S) = \{0.5S, 0.5 + 0.4(S - 0.2)\}.$$

The candidate values are $(0, 0.42)$ and $(0.5, 0.82)$; setting the CDF to zero below the first support point and to one at and above two completes each distribution. For $K_h = v_h = \mu_h = 1$, enumeration of (S, Y, R) gives

$$\Sigma_\lambda = \begin{pmatrix} .09 & .05 \\ .05 & .25 \end{pmatrix} + \lambda \begin{pmatrix} -.072 & -.0288 \\ -.0288 & 0 \end{pmatrix} + \lambda^2 \begin{pmatrix} .036 & .0288 \\ .0288 & .02304 \end{pmatrix}. \quad (7)$$

Proposition 1 (Integrated variance and band width separate). *The integrated-variance minimizer for the equal-weight grid is $\lambda_{IV} = 25/41$. It decreases $\text{tr } \Sigma_\lambda$ but increases the 95% critical value in (6) relative to the anchor.*

Proof. The trace is $.34 - .072\lambda + .05904\lambda^2$, whose minimizer is $25/41$. At the anchor, the union bound gives

$$\mathbb{P}(\|G_0\|_\infty > .99) \leq 2\bar{\Phi}(.99/.3) + 2\bar{\Phi}(.99/.5) = .04867037697 < .05.$$

At λ_{IV} , the second marginal variance is $.25856632957$, so

$$\mathbb{P}(\|G_{\lambda_{IV}}\|_\infty \leq .99) \leq 2\Phi\{.99/\sqrt{.25856632957}\} - 1 = .94845626635 < .95.$$

Thus $c_\Omega(0) < .99 < c_\Omega(\lambda_{IV})$. The inequalities have margins far exceeding the numerical integration tolerance used in the attached verification. $\square$

Deterministic bivariate normal integration gives the following additional values. The band-minimizing weight is a numerical optimizer; it is not needed for Proposition 1.

Rule	Weight	Trace	95% critical value
Anchor	0.000000	0.340000	0.983562
Integrated variance	0.609756	0.318049	0.996761
Band critical value	0.114860	0.332509	0.982639
Full	1.000000	0.327040	1.024169

The example demonstrates separation, not a large effect size for the new optimizer. Taking an independent continuous X embeds the same covariance geometry into a local conditional problem, up to a common positive scale.

4 Paired certification over a common confidence set

Let $\mathcal{C}_G$ be a nonempty compact confidence set of positive semidefinite joint matrices based on the selection sample, satisfying

$$\mathbb{P}(\Omega \in \mathcal{C}_G \mid \text{training}) \geq 1 - \delta. \quad (8)$$

Define the paired lower bound and borrowing rule by

$$\underline{g}_G(\lambda) = \inf_{Q \in \mathcal{C}_G} \{c_Q(0) - c_Q(\lambda)\}, \quad \hat{\lambda} \in \arg \max_{\lambda \in \Lambda} \underline{g}_G(\lambda), \quad (9)$$

where $0 \in \Lambda \subseteq [0, 1]$. A finite numerical grid always permits a maximizer. For a compact continuum, a measurable maximizer is used when it exists. If the maximum is zero, choose zero. The same Q must be used for the two critical values inside the braces. In particular, $\underline{g}_G(0) = 0$ exactly.

Theorem 2 (Certification and an oracle inequality). *Let $\lambda^* \in \arg \max_{\lambda \in \Lambda} g_\Omega(\lambda)$ and define, on the event $\Omega \in \mathcal{C}_G$,*

$$r_G(\lambda) = \sup_{Q \in \mathcal{C}_G} |g_Q(\lambda) - g_\Omega(\lambda)|.$$

The rule (9) satisfies

$$\mathbb{P}\{g_\Omega(\hat{\lambda}) < 0 \mid \text{training}\} \leq \delta, \quad (10)$$

$$0 \leq g_\Omega(\lambda^*) - g_\Omega(\hat{\lambda}) \leq r_G(\lambda^*) \quad \text{on } \{\Omega \in \mathcal{C}_G\}. \quad (11)$$

No uniqueness or curvature of the minimizer is required.

Proof. On the confidence event,

$$g_\Omega(\hat{\lambda}) \geq \underline{g}_G(\hat{\lambda}) \geq \underline{g}_G(0) = 0,$$

which proves (10). Also,

$$g_\Omega(\hat{\lambda}) \geq \underline{g}_G(\hat{\lambda}) \geq \underline{g}_G(\lambda^*) \geq g_\Omega(\lambda^*) - r_G(\lambda^*).$$

Optimality of λ^* gives the other side of (11). $\square$

A computationally certified lower bound may replace the exact infimum. An ordinary local optimizer for the inner minimization supplies an *upper* bound on that infimum, so its output alone is not a valid safety certificate. Analytic envelopes, globally valid bounds, or a rigorously controlled approximation error are required.

5 The paired error vanishes at the anchor

This section provides a fixed-grid refinement of Theorem 2. Let

$$\mathcal{K}_{\kappa, B} = \{Q \succeq 0 : \|Q\|_{\text{op}} \leq B, \quad H_\lambda Q H_\lambda^\top \succeq \kappa I_J \text{ for every } \lambda \in [0, 1]\}.$$

This is a convex compact class. The joint matrix Q itself may be singular: a zero correction block is allowed if the anchor covariance is positive definite.

Lemma 3 (Smoothness of the Gaussian rectangle quantile). *For fixed J and $\alpha \in (0, 1)$, the map*

$$q_\alpha(\Sigma) = \text{Quantile}_{1-\alpha} \|N(0, \Sigma)\|_\infty$$

is twice continuously differentiable on the positive definite cone. Its first and second derivatives are bounded, in the relevant induced operator norms, on $\{\kappa I_J \preceq \Sigma \preceq 2B I_J\}$.

Proof. Let $H(t, \Sigma) = \int_{[-t, t]^J} \varphi_\Sigma(z) dz$ for $t > 0$. Smoothness in Σ follows by differentiation under the integral on compact positive definite sets. Writing the integral over $[-1, 1]^J$ after the substitution $z = tu$ also establishes smoothness in t . The derivative $\partial_t H$ is a sum of integrals of strictly positive Gaussian densities over the faces of the rectangle and is strictly positive. The quantile is the unique positive solution of $H(t, \Sigma) = 1 - \alpha$. The implicit function theorem gives smoothness. Continuity and compactness give finite bounds on the derivatives. For reference, the first directional derivative is

$$Dq_\alpha(\Sigma)[E] = -\frac{\frac{1}{2} \int_{[-q, q]^J} \varphi_\Sigma(z) \{z^\top \Sigma^{-1} E \Sigma^{-1} z - \text{tr}(\Sigma^{-1} E)\} dz}{\partial_t H(q, \Sigma)}, \quad q = q_\alpha(\Sigma).$$

The constants can deteriorate with J or with $\kappa \downarrow 0$; the statement does not claim a growing-grid bound. $\square$

Theorem 4 (Anchor-adaptive perturbation and regret). *There is a finite constant $C = C(J, \alpha, \kappa, B)$ such that, for $Q, Q' \in \mathcal{K}_{\kappa, B}$,*

$$|g_Q(\lambda) - g_{Q'}(\lambda)| \leq C\lambda \|Q - Q'\|_{\text{op}}, \quad 0 \leq \lambda \leq 1. \quad (12)$$

In particular, suppose $\mathcal{C}_G \subseteq \mathcal{K}_{\kappa, B}$ and every member of $\mathcal{C}_G$ is within ε_G of a common center. On the confidence event,

$$0 \leq c_\Omega(\hat{\lambda}) - \inf_{\lambda \in \Lambda} c_\Omega(\lambda) \leq 2C\lambda^* \varepsilon_G. \quad (13)$$

Proof. Let M_1, M_2 bound the first and second derivatives in Lemma 3. Put $f_\lambda(Q) = q_\alpha(H_\lambda Q H_\lambda^\top)$. For any symmetric perturbation E ,

$$Df_\lambda(Q)[E] = Dq_\alpha(\Sigma_\lambda)[H_\lambda E H_\lambda^\top].$$

Since $\|H_\lambda\|_{\text{op}}^2 \leq 2$,

$$\begin{aligned} \|H_\lambda E H_\lambda^\top - H_0 E H_0^\top\|_{\text{op}} &\leq 3\lambda \|E\|_{\text{op}}, \\ \|\Sigma_\lambda - \Sigma_0\|_{\text{op}} &\leq 3B\lambda. \end{aligned}$$

Consequently,

$$|D\{f_0 - f_\lambda\}(Q)[E]| \leq (3M_1 + 6BM_2)\lambda \|E\|_{\text{op}}.$$

Integrate this derivative along the line segment from Q to Q' , which remains in $\mathcal{K}_{\kappa, B}$. This proves (12) with $C = 3M_1 + 6BM_2$. On the confidence event the set diameter is at most $2\varepsilon_G$, so Theorem 2 yields (13). $\square$

If an analytic bound uses $\hat{g}(\lambda) - C\lambda\varepsilon_G$ and its maximum is attained to tolerance τ , choosing the anchor whenever the attained lower bound is negative preserves safety; the corresponding regret is at most $2C\lambda^*\varepsilon_G + \tau$. Separate absolute critical-value intervals generally lose the factor λ . That loss matters when the beneficial weight approaches zero.

6 A fully observable covariance confidence ball

Theorem 2 does not rely on a confidence set supplied by an oracle. Here is one explicit construction, implemented in `verify_core.py`. It uses deliberately conservative constants; efficient multiplier calibration remains a separate numerical development.

Let the selection sample have size $n = n_G \geq 2$. Assume $0 \leq K_h \leq K_{\max}$ and $\pi \geq pc_e$. All statements are conditional on the independent training sample. Set $L = 2J + 1$ and define

$$X_i = \sqrt{p/v_h} (K_i, K_i U_{0,i}^\top, K_i B_i^\top)^\top, \quad Q = \mathbb{E} X_i X_i^\top, \quad \hat{Q} = n^{-1} \sum_i X_i X_i^\top.$$

For $f \in [0, 1]^J$, let

$$A_f = \begin{pmatrix} -f & I_J & 0 \\ 0 & 0 & I_J \end{pmatrix}.$$

Then $\Omega = A_{F_h} Q A_{F_h}^\top$. Use

$$\hat{\mu} = \mathbb{P}_G K_h, \quad \hat{f}_j = \left[\frac{\mathbb{P}_G K_h U_{0,y_j}}{\hat{\mu}} \right]_{[0,1]}, \quad \hat{\Omega} = A_{\hat{f}} \hat{Q} A_{\hat{f}}^\top.$$

The last matrix is automatically positive semidefinite.

Split δ into four positive parts $\delta_K, \delta_Q, \delta_N, \delta_\mu$. Let s_+ be an exact one-sided Clopper–Pearson upper confidence bound, at error δ_K , on $s = \mathbb{P}(K_h > 0)$, computed from the number of nonzero kernels. Let

$$t_Q = \log(2L^2/\delta_Q), \quad t_N = \log(2J/\delta_N), \quad t_\mu = \log(2/\delta_\mu).$$

Define

$$\begin{aligned} M_X &= \frac{K_{\max}}{c_e \sqrt{pv_h}}, & V_Q^+ &= \frac{2K_{\max}^4 s_+}{pc_e^3 v_h^2}, \\ V_N^+ &= \frac{2K_{\max}^2 s_+}{pc_e}, & V_\mu^+ &= K_{\max}^2 s_+, \\ r_Q &= \sqrt{2V_Q^+ t_Q/n} + \frac{4M_X^2 t_Q}{3n}, \\ r_N &= \sqrt{2V_N^+ t_N/n} + \frac{4K_{\max} t_N}{3npc_e}, \\ r_\mu &= \sqrt{2V_\mu^+ t_\mu/n} + \frac{2K_{\max} t_\mu}{3n}. \end{aligned}$$

For $\hat{\mu} > 0$, put $\zeta = \min\{1, (r_N + r_\mu)/\hat{\mu}\}$, and otherwise set $\hat{f}_j = 1/2$, $\zeta = 1$. Define

$$\varepsilon_G = (J+1)Lr_Q + 2\sqrt{J(J+1)} \zeta \{\|\hat{Q}\|_{\text{op}} + Lr_Q\}. \quad (14)$$

Proposition 5 (Observable confidence ball). *Under the stated boundedness, known-propensity, and sample-separation conditions,*

$$\mathbb{P}\{\|\hat{\Omega} - \Omega\|_{\text{op}} \leq \varepsilon_G \mid \text{training}\} \geq 1 - \delta.$$

For fixed J , bounded localization density, $\mu_h \asymp v_h$, and $n_G p v_h \rightarrow \infty$, the radius has order

$$\varepsilon_G = O_{\mathbb{P}} \left\{ \sqrt{\frac{\log(J/\delta)}{N_G}} + \frac{\log(J/\delta)}{N_G} \right\}, \quad N_G = n_G p v_h, \quad (15)$$

with dimension-dependent constants. The candidates need not converge to the true regressions for this fitted-path confidence statement.

Proof. When $R = 0$, $|U_0|, |B| \leq 1$; when $R = 1$, both are at most $1/\pi$. Thus every raw scaled coordinate is bounded by M_X , and

$$\mathbb{E}\|X_i\|_\infty^4 \leq \frac{2K_{\max}^4 s}{pc_e^3 v_h^2}.$$

For an entry of $X_i X_i^\top$, the centered absolute bound is $2M_X^2$. Bernstein's inequality and a union bound over L^2 entries yield $\|\hat{Q} - Q\|_{\max} \leq r_Q$ on an event of error at most $\delta_Q + \delta_K$, using the event $s \leq s_+$. The same reasoning gives simultaneous bounds r_N on the numerator means and r_μ on the kernel mean, with the second-moment bounds displayed above. These statements can be intersected even though the upper support bound and moment estimates use the same sample: the inequalities with the true support probability are intersected with $s \leq s_+$ before substitution.

Writing $\mathbb{E}(K_h U_{0,y_j}) = \mu_h F_h(y_j)$ gives

$$\left| \frac{\mathbb{P}_G K_h U_{0,y_j}}{\hat{\mu}} - F_h(y_j) \right| \leq \frac{r_N + r_\mu}{\hat{\mu}}.$$

Clipping to $[0, 1]$ cannot increase this error. Hence $\|\hat{f} - F_h\|_\infty \leq \zeta$. Now $\|A_f\|_{\text{op}} \leq \sqrt{J+1}$, $\|A_{\hat{f}} - A_{F_h}\|_{\text{op}} \leq \sqrt{J}\zeta$, and $\|\hat{Q} - Q\|_{\text{op}} \leq Lr_Q$. Expanding

$$A_{\hat{f}} \hat{Q} A_{\hat{f}}^\top - A_{F_h} Q A_{F_h}^\top$$

into a $\hat{Q} - Q$ term and two $A_{\hat{f}} - A_{F_h}$ terms gives (14), using $\|Q\|_{\text{op}} \leq \|\hat{Q}\|_{\text{op}} + Lr_Q$. The union error is at most the four allocated errors. Under localization, $s_+ = O_{\mathbb{P}}(v_h)$, $\hat{\mu} \asymp_{\mathbb{P}} v_h$, and $\|\hat{Q}\|_{\text{op}} = O_{\mathbb{P}}(1)$ for fixed J , which gives (15). $\square$

The positive semidefinite operator-norm ball supplied by Proposition 5 is sufficient for Theorem 2. Intersecting it with $\mathcal{K}_{\kappa,B}$ invokes additional, explicitly stated covariance conditions for Theorem 4; those conditions must not be silently imposed to exclude inconvenient data.

7 Weak borrowing: an information bound and a matching certificate

Consider a binary submodel with $S \in \{-1, 1\}$ equally likely and

$$Z = \mathbf{1}\{Y = 0\}, \quad \mathbb{P}_\theta(Z = 1 \mid S) = 1/2 + \theta S, \quad |\theta| < 1/4.$$

The marginal target is $1/2$. Let $q = 1/2$, $D = dS$ with fixed $0 < d < 1/4$, and $R \sim \text{Bernoulli}(p)$ independently. This is a valid binary-CDF augmentation family. Direct calculation gives

$$\sigma_\theta^2(\lambda) = p \text{Var}_\theta(U_\lambda) = \frac{1}{4} + (1-p)(d^2 \lambda^2 - 2d\theta\lambda). \quad (16)$$

For $0 < \theta < d$, the oracle weight is $\lambda^* = \theta/d$, and the Gaussian critical-value improvement is

$$g_\theta^* = z_{1-\alpha/2} \left\{ \frac{1}{2} - \sqrt{\frac{1}{4} - (1-p)\theta^2} \right\} = z_{1-\alpha/2}(1-p)\theta^2 + O(\theta^4). \quad (17)$$

For $\theta \leq 0$, no positive weight improves the anchor.

Theorem 6 (A local information obstruction). *Suppose a rule issues a positive-gain certificate on an event $\mathcal{A}$, with $\mathbb{P}_0(\mathcal{A}) \leq \delta$. From n_G observations in the binary model, let $N_G = n_G p$. Then*

$$\mathbb{P}_\theta(\mathcal{A}) \leq \delta + \sqrt{2N_G}|\theta|. \quad (18)$$

In particular, if $\theta = o(N_G^{-1/2})$, its probability of certifying the positive improvement is at most $\delta + o(1)$, while the optimal critical-value gain is $o(N_G^{-1})$.

Proof. The laws of (S, R) are identical under θ and zero. Conditional on verification, the Bernoulli divergence to probability $1/2$ is at most its chi-squared divergence, $4\theta^2$. Thus the observed-data divergence satisfies

$$\text{KL}(P_\theta^{n_G}, P_0^{n_G}) \leq 4n_G p \theta^2.$$

Pinsker's inequality gives total variation at most $\sqrt{2N_G}|\theta|$. Apply the event inequality and then (17). To embed the construction locally, take X independent of S , use a box kernel with support probability $r_h \asymp h^d$, and introduce the association θ only inside that support. All broadly observed variables have the same law under the two alternatives. The divergence bound replaces N_G by $n_G p r_h$, the expected local verified count. $\square$

Proposition 7 (Matching submodel certificate). *Conditional on $k \geq 1$ verified observations in the target neighborhood, let*

$$\hat{\theta} = \frac{1}{2k} \sum_{i=1}^k S_i(2Z_i - 1), \quad \theta_L = \hat{\theta} - \sqrt{\frac{\log(1/\delta)}{2k}}, \quad \hat{\lambda} = \min\{1, (\theta_L)_+/d\}.$$

The probability of selecting a strictly harmful positive weight is at most δ . For $\theta > 0$, the adoption probability is at least $1 - \beta$ whenever

$$\theta \geq \sqrt{\frac{\log(1/\delta)}{2k}} + \sqrt{\frac{\log(1/\beta)}{2k}}. \quad (19)$$

For $k = 0$, use $\hat{\lambda} = 0$.

Proof. Each summand divided by two lies in $[-1/2, 1/2]$ and has mean θ . One-sided Hoeffding bounds give $\mathbb{P}(\theta_L > \theta) \leq \delta$ and the adoption bound in (19). On the confidence event, if $\hat{\lambda} > 0$ then $0 < d\hat{\lambda} \leq \theta_L \leq \theta$. Therefore

$$\sigma_\theta^2(\hat{\lambda}) - \sigma_\theta^2(0) = (1 - p)d\hat{\lambda}(d\hat{\lambda} - 2\theta) \leq -(1 - p)(d\hat{\lambda})^2 < 0.$$

The Gaussian critical value is strictly increasing in the scalar variance, completing the proof. $\square$

These results identify a fixed-submodel signal scale and its squared gain scale. They do not assert that every possible multi-threshold covariance geometry has the same detection boundary. They also concern certification of a specified estimator family, not a lower bound for estimating an otherwise known marginal target over this submodel. With $\varepsilon_G \asymp N_G^{-1/2}$ and $\lambda^* \asymp \theta$, Theorem 4 gives a regret scale $\theta N_G^{-1/2}$, consistent with the submodel boundary. Using a constant-in- λ error bound would instead demand a gain of order $N_G^{-1/2}$ near the anchor and discard this adaptation.

8 Selection and final simultaneous coverage

A selection certificate and a coverage statement control different events. Keep training, selection, and final evaluation samples disjoint. No convergence of $\hat{\lambda}$ to a unique deterministic minimizer is needed for the following transfer lemma.

Theorem 8 (Protected-band construction). *Suppose (8) holds and, conditional on training, the evaluation sample satisfies*

$$\sup_{\lambda \in \Lambda} \sup_{t \in \mathbb{R}} \left| \mathbb{P} \left(\left\| n_E^{-1/2} \sum_{i \in E} H_\lambda \Psi(O_i) \right\|_\infty \leq t \middle| \text{training} \right) - \mathbb{P}(\|G_\lambda\|_\infty \leq t) \right| \leq \eta_E.$$

Let $U_G(\lambda) = \sup_{Q \in \mathcal{C}_G} c_Q(\lambda)$, and use (9). Then the grid band

$$\hat{F}_{E, \hat{\lambda}}(y_j) \pm \frac{U_G(\hat{\lambda})}{\hat{f}_{h,E} \sqrt{n_E p v_h}}$$

has simultaneous coverage at least $1 - \alpha - \delta - \eta_E$. If the kernel denominator is zero, report the trivial $[0, 1]$ band instead. Moreover, its untruncated common half-width is no greater than that of the correspondingly protected anchor band using $U_G(0)$ and the same sample.

Proof. On the confidence event, $U_G(\hat{\lambda}) \geq c_\Omega(\hat{\lambda})$. Given training and selection, the selected weight is fixed and the evaluation sample is independent. The approximation hypothesis gives conditional grid coverage at least $1 - \alpha - \eta_E$, using the exact identity (5). Integration and the confidence-set error give the asserted bound. For every $Q \in \mathcal{C}_G$, optimality in (9) implies $g_Q(\hat{\lambda}) \geq \underline{g}_G(\hat{\lambda}) \geq 0$. Hence $c_Q(\hat{\lambda}) \leq c_Q(0)$ for every such Q , and taking suprema yields $U_G(\hat{\lambda}) \leq U_G(0)$. $\square$

The width comparison is with an anchor given the *same* covariance protection, not with an uninflated conventional anchor. A desired total noncoverage probability must allocate the covariance-certificate and Gaussian-approximation errors as well as α . Alternatively, a uniformly valid independently calibrated evaluation-fold multiplier band gives marginal coverage $1 - \alpha + o(1)$ irrespective of whether the selection certificate succeeds; its realized width ordering is then an empirical question. The joint event of coverage and true Gaussian gain has probability at least $1 - \alpha - \delta - o(1)$. Gaussian comparison, anti-concentration, and high-dimensional rectangle approximation are appropriate tools for controlling the remaining approximation terms (Chernozhukov et al., 2015, 2016).

The original rotating three-role estimator is not automatically covered by the above conditional argument: conditioning on all rotating selections can reveal information about several evaluation blocks. A replacement result must explicitly control this dependence or introduce an outer independent selection stage.

8.1 From a threshold grid to an entire CDF

Suppose simultaneous lower and upper limits l_j, u_j cover $F_h(y_j)$. Then

$$l(y) = \max\{0, \max_{y_j \leq y} l_j\}, \quad u(y) = \min\{1, \min_{y_j \geq y} u_j\}$$

cover $F_h(y)$ for every real y , with the natural empty-set conventions. The proof is monotonicity: each left grid value is a lower bound for the CDF to its right, and each right grid value is an

upper bound to its left. Intersecting limits with $[0, 1]$ preserves coverage. An inconsistent interval family off the coverage event can be replaced by $[0, 1]$. This extension provides whole-CDF coverage without a smooth density; it does not identify finite-grid width optimality with continuum-supremum optimality. A sharp growing-grid process result must account for interpolation error and deterioration of the fixed-grid perturbation constants.

9 Scientific target for the air-monitoring application

The supplied manuscript reports 2,814 collocated site-days from seven sites, local humidity targets of 30, 50, and 70 percent, and calendar-week assignment to fitting, selection, and evaluation roles (Section 5.1, pages 23–25). No new EPA data analysis is claimed in this note.

Let Y be reference $\text{PM}_{2.5}$ and $\tilde{S}$ a corrected sensor measurement obtained using a prespecified external correction or a separately trained, frozen calibration. The proposed scientific target is

$$\Delta_h(t; r) = \frac{\mathbb{E}[K_h(X - r)\{\mathbf{1}(Y \leq t) - \mathbf{1}(\tilde{S} \leq t)\}]}{\mathbb{E}K_h(X - r)}.$$

It equals sensor-minus-reference exceedance probability at the declared localization scale. Positive values indicate overstatement of exceedance frequency, not an individual-level false-positive rate. The paired estimator is

$$\hat{\Delta}_{h,\lambda}(t; r) = \frac{\mathbb{P}_E K_h(X - r)\{U_{\lambda,t} - \mathbf{1}(\tilde{S} \leq t)\}}{\mathbb{P}_E K_h(X - r)}.$$

Its reference block is

$$\Psi_{0,t}^\Delta = \sqrt{p/v_h} K_h(X - r)\{U_{0,t} - \mathbf{1}(\tilde{S} \leq t) - \Delta_h(t; r)\},$$

and its correction block is the same $\sqrt{p/v_h} K_h B_t$ as before. Thus the entire paired-certification construction applies to the actual scientific difference, retaining all sensor–reference covariances. The observable-radius constants in Section 6 must be recalculated for this $[-1, 1]$ -valued contrast; the $[0, 1]$ clipping used there is not applied to Δ_h . The borrowing weight chosen for the reference CDF need not minimize the band for this difference. The criterion should match the quantity ultimately reported.

The empirical outputs are concentration-threshold regions where the sign of Δ_h is resolved, simultaneous uncertainty at the declared humidity values, high-quantile calibration differences when tail support permits them, and reference-budget requirements for recovering these conclusions. The frozen correction, time aggregation, site weighting, threshold grid, and humidity targets must be defined before inspecting new outcome contrasts. Reference-informed cleaning cannot remove errors that would remain invisible under sparse verification. Calendar-week dependence requires cluster-level covariance contributions and a corresponding approximation theorem; the independent site-day count must not be substituted for the number of independent blocks. Joint claims across three humidity targets also require an explicit familywise error allocation or a joint process.

10 Implementation record and the next mathematical deliverable

The attached Python module was executed successfully. It performs exact finite-support covariance enumeration; deterministic Gaussian rectangle integration; independent marginal-bound checks for Proposition 1; perturbation checks of the paired anchor identity; exhaustive deterministic safety

checks for the binary certificate over possible observed counts; exact binomial adoption calculations; and one smoke test of the observable covariance-ball routine. The smoke test is not an empirical confidence-coverage study, and the perturbation checks are not substitutes for Theorem 4.

The completed mathematical objects in this note are the fixed-grid separation, the generic certificate and oracle inequality, the fixed-grid paired perturbation theorem, a concrete covariance confidence ball, the binary information bound and matching certificate, and the conditional coverage-transfer statement. The next deliverable is a sharp, computable paired-gain calibration for a growing threshold grid, including its numerical approximation error, and the dependence-aware connection to the manuscript’s estimation scheme. That development should preserve the local paired cancellation instead of reverting to two separate absolute critical-value intervals. The EPA reanalysis should then use the paired scientific target above. These steps are part of the existing strengthening route, not new research directions.

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